LESSON PLAN

HELICOPTER THEORY (11)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	26
Total No. of Lesson Plans	27	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Dynamic stability: longitudinal and lateral, Phugoid, Dutch roll, Fixed wing autorotation
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Dynamic stability: longitudinal and lateral, Phugoid, Dutch roll, Fixed wing autorotation. Short period oscillation, spiral divergence and Directional divergence.
5. Specific Objectives	11.1.4.1 Trainee will be able to describe aircraft stability in all the three axis

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of stability axes diagrams; Instructor supervision required for diagram interpretation
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) Define static stability in longitudinal axis.
(b) List the three axes of aircraft motion.
(c) Explain the difference between static and dynamic stability.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Longitudinal dynamic stability definition • Short period oscillation characteristics • Phugoid oscillation characteristics • Lateral dynamic stability definition • Dutch roll oscillation characteristics	L & Ds	PPT, IB	15 min
(b)	• Spiral divergence characteristics • Directional divergence characteristics • Fixed wing autorotation principles • Comparison of oscillatory modes • Stability criteria summary	L & Ds	PPT, PDF	20 min
(c)	• Application of stability concepts • Analysis of flight envelope limits • Review of dynamic modes • Instructor clarification of doubts • Preparation for next lesson	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Describe the nature of the Phugoid oscillation.
(b) Differentiate between Dutch roll and Spiral divergence.
(c) Explain the role of fixed wing autorotation in stability.

12. Home Assignment and Follow Up	Prepare a brief note on the relationship between longitudinal stability and the Short Period oscillation for the next lesson on Low speed wind tunnel testing.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Low speed wind tunnel and wind tunnel balances. Demonstration on pressure plotting experiments. Measurements of lift and drag in wind tunnel.	Refer to the master syllabus reference list	LP 27

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min